

# DP Computer Science: A 2.2.2 Servers (HL Only)

## 1. Core Concepts & Learning Objectives

### Learning Intentions

By the end of this lesson, students will be able to:

- Define the **role of a server** in a network
- Describe the functions of key server types: **DNS, DHCP, file, mail, proxy, web**
- Analyse how factors like **scalability, reliability, and security** influence server choice
- Apply knowledge to **recommend suitable server types** for different scenarios

### Key Vocabulary

| Term         | Definition                                                                                          |
|--------------|-----------------------------------------------------------------------------------------------------|
| Server       | A specialised computer that provides services to other computers (clients) on a network             |
| Client       | A device that requests services from a server                                                       |
| DNS Server   | Translates domain names (e.g., <a href="#">google.com</a> ) into IP addresses                       |
| DHCP Server  | Automatically assigns IP addresses to devices on a network                                          |
| File Server  | Stores and manages files for sharing across a network                                               |
| Mail Server  | Handles sending, receiving, and storing emails                                                      |
| Proxy Server | Acts as an intermediary between clients and the internet, filtering requests and improving security |
| Web Server   | Hosts websites and delivers web pages to users                                                      |
| Scalability  | The ability to handle increasing numbers of users or data                                           |
| Reliability  | The ability to operate without failure                                                              |
| Redundancy   | Having backup systems in case of failure                                                            |

### IB ATL Skills

| Skill           | Description                                                    |
|-----------------|----------------------------------------------------------------|
| Thinking        | Critical thinking, analysis, problem-solving                   |
| Communication   | Explaining concepts and collaborating in discussions           |
| Self-Management | Organisation, time management                                  |
| Research        | Gathering information, evaluating sources (optional extension) |

## 2. What is a Server?

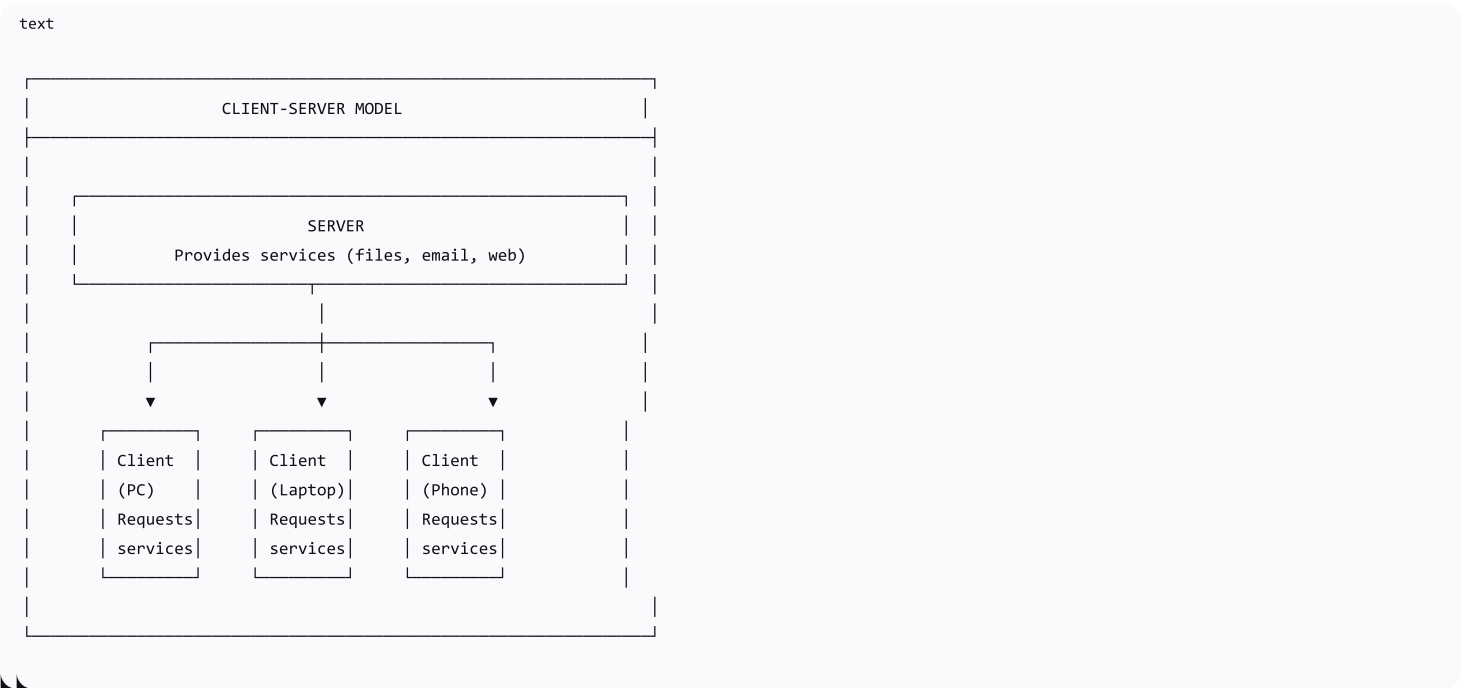
### Definition

A server is a specialised computer dedicated to providing specific services to other computers (clients) on a network.

Servers are designed to be:

- **Powerful** — handle many requests simultaneously
- **Reliable** — run continuously without failure
- **Secure** — protect data and control access

### Server vs. Client

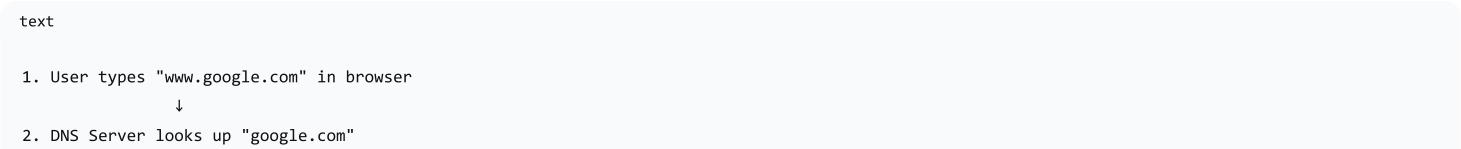


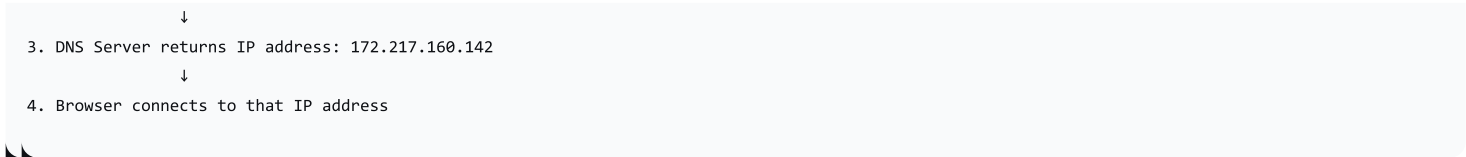
## 3. Types of Servers

### DNS Server (Domain Name System)

| Aspect         | Description                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------|
| Function       | Translates domain names (e.g., <a href="#">google.com</a> ) into IP addresses (e.g., 172.217.160.142)          |
| Role           | The "phonebook" of the internet                                                                                |
| Why it matters | Humans remember names; computers remember numbers                                                              |
| Example        | When you type " <a href="#">youtube.com</a> ", the DNS server finds its IP address so your browser can connect |

How DNS Works:





**Analogy:** A DNS server is like a **phonebook**—you look up a person’s name to find their phone number.

## DHCP Server (Dynamic Host Configuration Protocol)

| Aspect         | Description                                                                         |
|----------------|-------------------------------------------------------------------------------------|
| Function       | Automatically assigns IP addresses and network settings to devices                  |
| Role           | Simplifies network management; prevents IP address conflicts                        |
| Why it matters | Eliminates the need for manual configuration of every device                        |
| Example        | When you join a Wi-Fi network, DHCP assigns your device an IP address automatically |

What DHCP Provides:

| Information     | Description                                    |
|-----------------|------------------------------------------------|
| IP Address      | A unique address for the device on the network |
| Subnet Mask     | Defines the network segment                    |
| Default Gateway | IP address of the router                       |
| DNS Server      | IP address for domain name resolution          |

**Analogy:** A DHCP server is like a **reception desk**—when a guest arrives, they are assigned a room number (IP address) and given necessary information.

## File Server

| Aspect         | Description                                                      |
|----------------|------------------------------------------------------------------|
| Function       | Stores, manages, and shares files across a network               |
| Role           | Centralised file storage for organisations                       |
| Why it matters | Enables collaboration; provides central backup; controls access  |
| Examples       | School network drive, Google Drive, Dropbox (cloud file servers) |

Key Features:

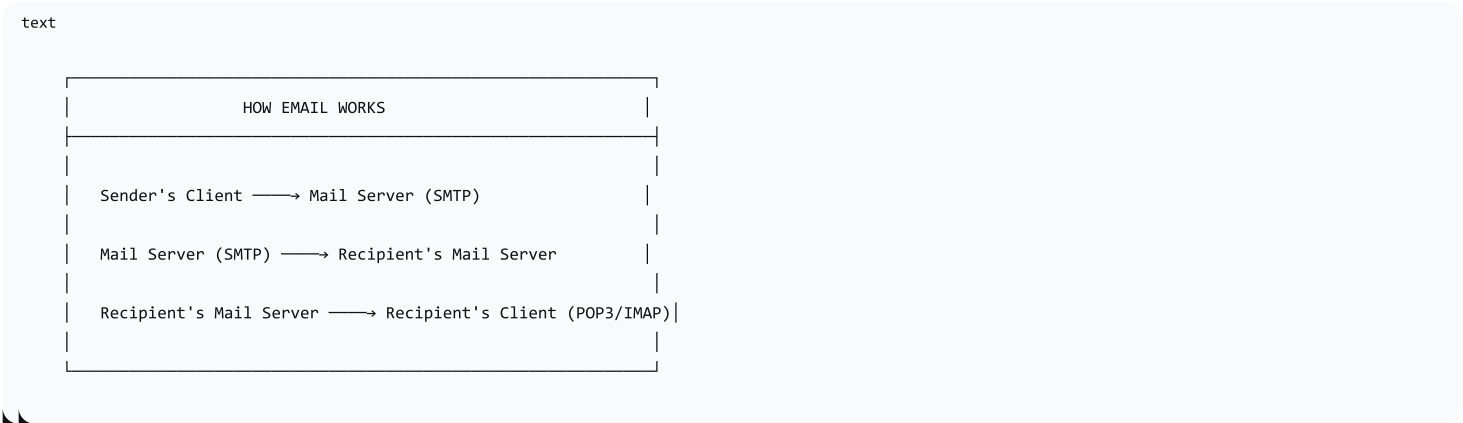
- Centralised storage
- Access control (who can read/write files)
- Backup and recovery
- File sharing across multiple users

**Analogy:** A file server is like a **library**—people can borrow and return books (files), and the librarian (server) manages everything.

## Mail Server

| Aspect         | Description                                                        |
|----------------|--------------------------------------------------------------------|
| Function       | Handles sending, receiving, and storing emails                     |
| Role           | Manages email communication for an organisation                    |
| Why it matters | Enables reliable internal and external email communication         |
| Examples       | Microsoft Exchange, Gmail servers, private corporate email servers |

How Email Works:



Protocols Used:

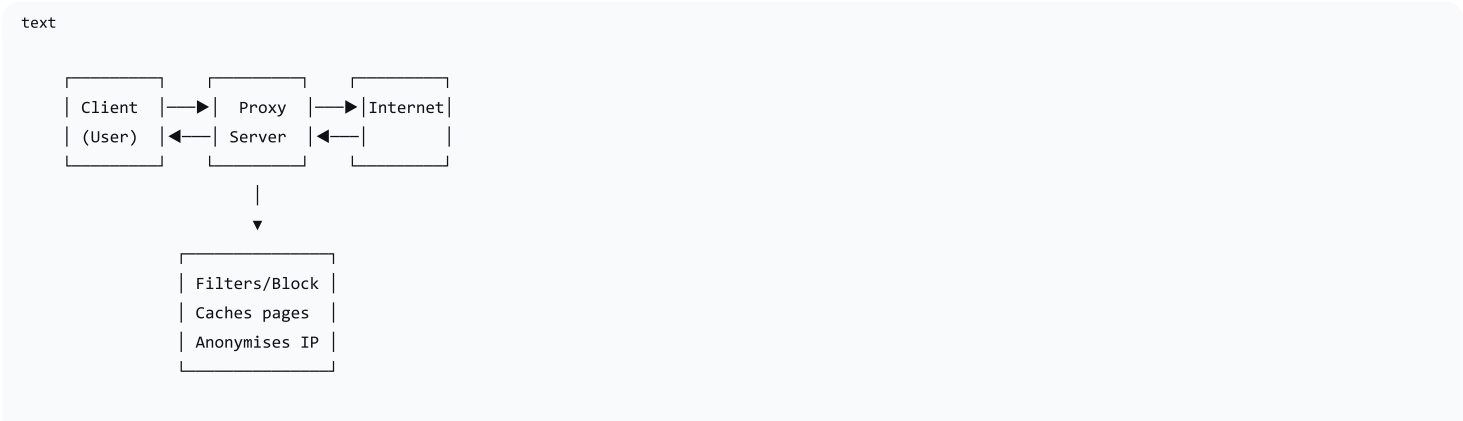
- SMTP — Sending emails
- POP3 / IMAP — Receiving emails

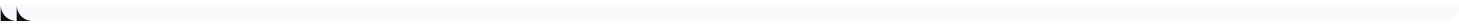
Analogy: A mail server is like a **post office**—it receives, sorts, and delivers mail (email) to the right mailbox.

Proxy Server

| Aspect         | Description                                                              |
|----------------|--------------------------------------------------------------------------|
| Function       | Acts as an intermediary between clients and the internet                 |
| Role           | Filters requests, improves security, caches content                      |
| Why it matters | Enhances security; can block inappropriate content; improves performance |
| Examples       | School internet filtering, corporate web gateways                        |

How a Proxy Server Works:





Key Functions:

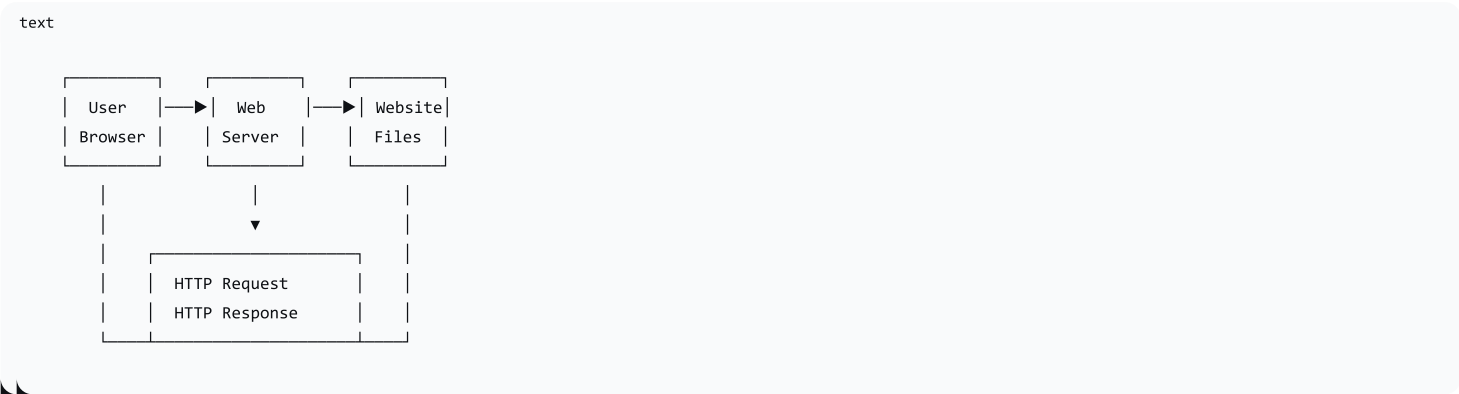
- **Content filtering** — blocks inappropriate or unsafe websites
- **Caching** — stores frequently accessed pages for faster loading
- **Anonymity** — hides user IP addresses
- **Access control** — restricts internet access based on policies

**Analogy:** A proxy server is like a **security guard** at a building entrance—they check who is allowed in, block unauthorised people, and keep a record of who comes and goes.

Web Server

| Aspect         | Description                                      |
|----------------|--------------------------------------------------|
| Function       | Hosts websites and delivers web pages to users   |
| Role           | Stores website files and serves them to browsers |
| Why it matters | Makes websites accessible to the world           |
| Examples       | Apache, Nginx, Microsoft IIS                     |

How a Web Server Works:



Key Features:

- Stores HTML, CSS, JavaScript, images, and videos
- Responds to HTTP/HTTPS requests from browsers
- Can handle multiple concurrent users
- Often supports databases for dynamic content

**Analogy:** A web server is like a **restaurant kitchen**—you order (request) a dish, and the kitchen (server) prepares and delivers it to your table (browser).

4. Server Types Summary Table

| Server Type | Function                                | Key Feature                 | Example Use                                               |
|-------------|-----------------------------------------|-----------------------------|-----------------------------------------------------------|
| DNS         | Translates domain names to IP addresses | "Phonebook of the internet" | Accessing <a href="https://www.google.com">google.com</a> |
| DHCP        | Automatically assigns IP addresses      | Simplifies network setup    | Connecting to Wi-Fi                                       |
| File        | Stores and shares files centrally       | Centralised storage         | School network drives                                     |
| Mail        | Handles email sending and receiving     | Email management            | Corporate email                                           |

| Server Type | Function                                  | Key Feature          | Example Use               |
|-------------|-------------------------------------------|----------------------|---------------------------|
| Proxy       | Intermediary between clients and internet | Security and caching | School internet filtering |
| Web         | Hosts websites and delivers web pages     | Website hosting      | Hosting a website         |

## 5. Factors Influencing Server Choice

| Factor             | Description                                    | Considerations                                                 |
|--------------------|------------------------------------------------|----------------------------------------------------------------|
| Scalability        | Ability to handle growth                       | Will the server need to support more users/requests over time? |
| Reliability        | Ability to operate without failure             | What is the acceptable downtime? Is redundancy needed?         |
| Security           | Protection of data and systems                 | What data is being stored? What are the security requirements? |
| Cost               | Budget for hardware, software, and maintenance | What is the initial and ongoing cost?                          |
| Performance        | Speed and processing power                     | How many requests per second must be handled?                  |
| Ease of Management | Administrative complexity                      | What technical expertise is required?                          |

## Reliability and Redundancy

| Term            | Description                                                          |
|-----------------|----------------------------------------------------------------------|
| Reliability     | The ability of a server to operate without failure                   |
| Redundancy      | Having backup systems (e.g., multiple servers, power supplies)       |
| Uptime          | The percentage of time a server is operational (e.g., 99.99% uptime) |
| Fault Tolerance | The ability to continue operating when components fail               |

Example: A government agency handling sensitive data prioritises **security** and **reliability** over cost.

## 6. Scenario-Based Activity

### Instructions

For each scenario, identify which server types are needed and justify your choice.

### Scenario 1: Small Business Network

**Problem:** A small business with 15 employees needs a network to share files, access the internet, and use a common printer. They want data backed up regularly and protected from unauthorised access.

| Server Type  | Needed?         | Why?                                                   |
|--------------|-----------------|--------------------------------------------------------|
| File Server  | ✔               | Essential for centralised file storage and sharing     |
| DHCP Server  | ✔               | Automatically assigns IP addresses to employee devices |
| Proxy Server | ✔ (Optional)    | Enhances security; filters internet traffic            |
| Mail Server  | ✗ (or optional) | Could use third-party provider like Gmail instead      |
| Web Server   | ✗               | Not needed for internal business network               |

### Scenario 2: Government Agency Data Security

**Problem:** A government agency handles highly sensitive citizen data. They need secure storage and protection from unauthorised access.

| Server Type  | Needed?      | Why?                                           |
|--------------|--------------|------------------------------------------------|
| File Server  | ✔            | Securely store and manage sensitive data       |
| Proxy Server | ✔            | Filter traffic and prevent unauthorised access |
| DNS Server   | ✔ (Likely)   | Internal name resolution within the agency     |
| Mail Server  | ✔ (Optional) | Internal secure email communication            |

**Key Focus:** Security, strong access controls, encryption, backup mechanisms.

### Scenario 3: School Club Website

**Problem:** A school club wants to create a website to share information, post announcements, and allow members to sign up for events.

| Server Type | Needed?      | Why?                                                |
|-------------|--------------|-----------------------------------------------------|
| Web Server  | ✔            | Essential to host the website and deliver web pages |
| DNS Server  | ✔ (Optional) | If the club has its own domain name                 |
| File Server | ✗            | Not required for a simple website                   |

### Scenario 4: Online Gaming Community

**Problem:** A group wants to create a platform for their online gaming community. They need to host game servers and handle high traffic loads.

| Server Type  | Needed?      | Why?                                               |
|--------------|--------------|----------------------------------------------------|
| Game Servers | ✔            | Specialised servers to host and run online games   |
| File Server  | ✔            | Store game data, user profiles, and files          |
| Web Server   | ✔            | Host a web portal for the community                |
| DNS Server   | ✔ (Optional) | Help players connect using a memorable domain name |

**Key Focus:** High traffic handling, low latency, reliability.

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## 7. Lesson Sequence

### Lesson Plan

| Phase                | Time   | Activity                                                                                                  |
|----------------------|--------|-----------------------------------------------------------------------------------------------------------|
| Engage               | 10 min | "When you check email or browse a website, what's happening behind the scenes?"; introduce server concept |
| Explore              | 25 min | Explore key server types (DNS, DHCP, file, mail, proxy, web); use diagrams and examples                   |
| Interactive Activity | 15 min | Think-Pair-Share: match server types to scenarios and justify choices                                     |
| Wrap-up              | 10 min | Factors influencing server choice; emerging trends; real-world connection                                 |

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### Activity: Server Matching

Instructions: Match each server type to its function and real-world example.

| Server Type | Function                                | Real-World Example                                        |
|-------------|-----------------------------------------|-----------------------------------------------------------|
| DNS         | Translates domain names to IP addresses | Accessing <a href="https://www.google.com">google.com</a> |
| DHCP        | Assigns IP addresses automatically      | Joining a Wi-Fi network                                   |
| File        | Stores and shares files centrally       | School network drive                                      |
| Mail        | Handles email communication             | Corporate email system                                    |
| Proxy       | Filters internet traffic                | School internet filtering                                 |
| Web         | Hosts websites                          | Hosting a club website                                    |

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## 8. Assessment Activities

| Assessment Type          | Description                                         |
|--------------------------|-----------------------------------------------------|
| Class Participation      | Observe engagement and contributions                |
| Server Matching Activity | Match server types to functions and scenarios       |
| Scenario-Based Questions | Recommend suitable server types and justify choices |
| Quiz                     | Multiple-choice quiz on eBook page A2.2.2           |

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#### Sample Quiz Questions:

1. Which server translates domain names to IP addresses?
2. What is the primary function of a DHCP server?
3. Which server would you use to host a school website?
4. Which server type enhances security by acting as an intermediary?



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## 9. Differentiation

| Level     | Strategy                                                                                                          |
|-----------|-------------------------------------------------------------------------------------------------------------------|
| Support   | Provide simplified diagrams, explanations, sentence starters, graphic organisers; pair students strategically     |
| Challenge | Encourage research on real-world case studies; explore advanced topics like server virtualisation, cloud security |

Optional Extension: Students research a specific server type or emerging trend and present findings to the class.

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## 10. Resources

- **Presentation** (Slide Deck)
  - **eBook** (A2.2.2 with quiz)
  - **Worksheet:** Server matching + scenario-based questions
  - **Quizlet** (A2.2.2 vocabulary flashcards)
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## 11. Summary: Key Takeaways

| Server | Key Function                                       |
|--------|----------------------------------------------------|
| DNS    | Translates domain names → IP addresses (phonebook) |
| DHCP   | Automatically assigns IP addresses                 |
| File   | Centralised file storage and sharing               |
| Mail   | Handles email communication                        |
| Proxy  | Filters traffic; improves security                 |
| Web    | Hosts websites and delivers web pages              |

| "Servers are the backbone of modern networks—they provide the services that make the internet and organisations function."